

CompSoc Challenge - AdaHack 2026

Goal

Use open source imaging and sensor data to identify environmental disasters and anomalies. Your solution should focus on finding events that are under-reported or easy-to-miss.

What to Deliver

Your submission should include:

- an explanation of what anomaly or event you're detecting;
- an overview of what data sources were used;
- the methods, predictions, and models used (we love to hear technical specifics!);
- what results and outcomes you were able to achieve;
- a quick demo showcasing the solution.

Your application could be a command line tool, a website, a graphical dashboard, or anything in between.

Judging Criterion (bonus points)

We're especially interested in seeing solutions which:

- leverage small-compute solutions that avoid the use of large models, especially those which apply classical methods such as statistics, clustering, and signal processing;
- make efficient use of pre-existing (non-language) models in novel ways;
- can be run on a regular Laptop or even Raspberry Pi;
- have verifiable results against pre-existing baseline metrics.

Examples

For some examples of what we're looking for, you could:

- use NOAA satellite imaging to identify deforestation;
- use BGS data to identify unreported small-scale earthquakes in remote regions;
- use the GUVI dataset to identify and predict auroral anomalies after geomagnetic storms.

These are just examples! We'd love to see what other data sources y'all can come up with.